

META ADS CAMPAIGN PERFORMANCE TRACKER

Product Requirements Document

Internal multi-tenant reporting & intelligence platform for Meta (Facebook) Ads — built for real estate marketing operations, architected for white-label expansion.

Prepared for: Easy — Founder

Document Version: 4.3 — ER diagram regenerated cleanly via PlantUML

Supersedes: Version 4.2 (August 5, 2026), Version 4.1 (August 4, 2026), and Version 1.0 (July 28, 2026)

Date: August 5, 2026 (rev. 2)

Table of Contents

Revision Note — v4.1

1. Executive Summary
2. Problem Statement
3. Objectives
4. Success Metrics
5. Assumptions & Constraints
6. Out of Scope / Non-Goals
7. User Roles
8. User Stories
9. Functional Requirements — Modules
10. Non-Functional Requirements
11. Dashboard Specifications
12. Screen-by-Screen Requirements
13. KPI Framework
14. Filters
15. Reports
16. Visualizations
17. Alerts & Notification Rules
18. Data Model & ER Diagram
19. Database Design
20. API Design
21. Role-Based Access Control (RBAC) Matrix
22. UI/UX Guidelines
23. Technical Architecture
24. MVP vs. Phase 2 Roadmap
25. Rollout / Migration Plan
26. Risks
27. Glossary
28. Acceptance Criteria (MVP)
29. Appendix

Revision Note — v4.2

This revision corrects Version 4.1, which removed the CEO/Business Owner and Marketing Director personas based on an outdated read of the codebase. The ER diagram supplied for this revision (current schema, migrations 0001–0010) shows the actual live role model, and it does not match either the v4.0 persona set or the v4.1 four-role guess. This version replaces both with what the schema actually implements.

What the schema actually does

- Roles are not a fixed enum. `workspace_member` links a user to a workspace via a `role_id`, and `role` is its own table: `id`, `workspace_id` (nullable), `name`, `is_system_template`.
- `workspace_id` = null on a role row means it's a system template available to every workspace. Five system templates exist today: CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator.
- Two additional templates — Supervisor and Campaign Executive — existed at some point and were removed in migration 0007.
- A permission table (`role_id`, `resource`, `action`) grants fine-grained permissions per role, which means a workspace can also define its own custom roles beyond the five templates, built from the same resource/action primitives.
- There is no 'Manager', 'Marketing Executive', or 'Viewer' role anywhere in the schema — those names, introduced in v4.1, do not exist in the build and every reference to them has been removed from this document.

Correction: v4.1 was built from an old conversation summary describing an early `workspace_role` enum (`admin/manager/marketing_executive/viewer`). That enum is not what's in the schema now — it was evidently superseded by the role/permission model shown in the supplied ER diagram. This version (v4.2) is built directly from that diagram instead of from conversation history, and should be treated as the more reliable source going forward.

Other changes in this revision

- Section 18 (Data Model & ER Diagram) has been rewritten entity-by-entity against the supplied diagram, including tables not previously documented: `role`, `permission`, `sync_log`, `audit_log`, `notification`, `historical_metrics`, and the workspace-level `jsonb` configuration columns.
- The actual ER diagram is embedded as a full-page visual at the end of Section 18, replacing the earlier placeholder.
- Section 19 (Database Design) and Section 21 (RBAC Matrix) are updated to describe the role/permission table structure rather than a fixed enum.

1. Executive Summary

This document specifies a purpose-built, internal reporting and analytics platform for monitoring Meta (Facebook) Ads campaign performance across one Business Manager and multiple Ad Accounts, initially serving a real estate marketing team of 10–25 people, and architected from day one as a multi-tenant (workspace-based) system so it can be white-labelled to other clients later.

The system is explicitly a monitoring, analysis, comparison, and reporting tool — it does not create, edit, or manage live ad campaigns. It becomes the single source of truth that answers, at a glance, what the team actually asks every day: where money is going, which campaigns are working, which need attention, and what to do next.

The platform is designed to be built in a focused MVP first — core campaign tracking, lead analytics, role-based dashboards, and a rule-based Campaign Profitability & Continue/Pause Advisor — with deeper analytics (predictive pacing, lead-quality scoring, natural-language AI narratives) deferred to later phases. This phased approach keeps the initial build scoped to what the team needs on day one, while the underlying architecture is designed to support the full feature set as it's phased in.

2. Problem Statement

Today, campaign performance data lives natively inside Meta Ads Manager, which is built for media buyers running campaigns — not for a small team asking business questions. There is no single view that connects ad spend to property or city performance, and no historical, comparative, or plain "is this working" reporting layer.

Core pain points this system resolves

- No unified view across 300+ campaigns spanning multiple ad accounts under one Business Manager.
- The team cannot quickly answer "what changed since yesterday" or "which campaign needs attention now."
- No systematic way to attribute spend and leads to property or city.
- No standardized daily/weekly/monthly reporting — reporting today is manual and inconsistent.
- No alerting — underperformance (CTR drop, budget exhaustion, learning-limited status) is discovered late, not proactively.
- No defined, repeatable way to tell whether a specific campaign is actually profitable and worth continuing — pause/continue calls are made on gut feel, not a threshold anyone agreed on in advance.
- Revenue/ROI data is partial (deal closures aren't always tracked against ad spend), so true ROAS is currently unmeasurable — the system needs to support an estimated-ROI model using assumed conversion rate and average deal value until full CRM integration exists.

3. Objectives

Each objective below states the business question it answers and why that matters — this is intentional: every module in this PRD traces back to one of these questions, so nothing is built without a clear business purpose.

Objective	Business Question Answered	Why It Matters
Real-time spend visibility	How much are we spending today, across all accounts?	Prevents budget overruns and surprises at month-end billing.
Profitability tracking	Which campaigns are profitable (or trending toward it)?	Redirects budget toward what works, away from what doesn't.

Objective	Business Question Answered	Why It Matters
Account health monitoring	Which ad accounts are underperforming?	Catches account-level issues (policy flags, low delivery) before they compound.
Attention triage	Which campaign needs immediate attention?	Turns a reactive team into a proactive one — issues are surfaced, not discovered.
Profitability & pause guidance	Is this campaign making or losing money, and should it keep running?	Turns raw spend/lead data into a direct continue-or-pause call, without requiring manual analysis every time.
ROI visibility	What is today's (estimated) ROI?	Connects ad spend to business outcomes, not just vanity ad metrics.
Property intelligence	Which property generated the most leads?	Informs where to invest marketing budget across the portfolio.
Geographic intelligence	Which city performs best?	Guides geographic expansion and city-level budget allocation.
Creative intelligence	Which creative performs best?	Reduces creative fatigue and guides the creative team's next brief.
Audience intelligence	Which audience performs best?	Improves targeting efficiency and lowers cost per lead over time.
Change detection	What changed compared to yesterday?	Day-over-day deltas are often more actionable than absolute numbers.
Budget action	Which campaigns should be paused or get more budget?	Turns analysis into a direct, actionable recommendation.

4. Success Metrics

How we'll know the MVP succeeded, 60–90 days after launch:

- 100% of active campaigns (300+) syncing daily with zero manual data entry.
- Daily/weekly reports generated and delivered automatically to each role, replacing manual reporting.
- Time-to-detect a critical issue (e.g. budget exhausted, CTR collapse) drops from "discovered days later" to under 2 hours via alerts.
- At least 80% of active users log in weekly without being reminded — indicating the dashboards are genuinely useful, not just built.
- The team can answer any of the 12 objective questions above in under 60 seconds using the dashboard, without asking a team member to pull a manual report.

5. Assumptions & Constraints

5.1 Assumptions

- One Facebook Business Manager houses multiple Ad Accounts at launch; additional Business Managers (for future white-label clients) will map to additional workspaces.
- Campaign volume at launch is 300+ active campaigns, scaling over time — the data model and sync jobs must be designed for this volume from day one, not retrofitted.

- Lead destination is mixed: some campaigns use native Meta Lead Forms (leads retrievable via Meta API), others send traffic to landing pages/websites (leads arrive via pixel/CRM webhook, not the Meta API directly). The system must support both ingestion paths.
- True revenue/ROAS data is only partially available. Until full CRM integration exists, ROI will be computed using an editable assumed conversion rate and average deal value per property/campaign type, clearly labeled as "Estimated ROI" everywhere it's shown — never presented as confirmed revenue.
- No strict campaign naming convention exists today, and the system will not attempt to auto-parse campaign names to extract property, city, or any other attribute — tagging is manual, in-app, done once per campaign by whoever sets it up.
- The team runs on a flat model: roles are defined via a role/permission table rather than a fixed set — five system-template roles exist today (CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator), all workspace-scoped rather than campaign-scoped — no user's view is restricted to a subset of campaigns within a workspace they belong to.

5.2 Constraints

- Subject to Meta Marketing API rate limits and data retention windows (Meta API insight data is generally reliable in detail for ~37 months, with some breakdowns limited further).
- Meta attribution windows and platform-reported metrics won't perfectly reconcile with CRM-reported leads/sales — the system should surface both and be transparent about the gap, not force a false reconciliation.
- WhatsApp Business API (for alerting) requires an approved WhatsApp Business Account and message templates pre-approved by Meta — this is a dependency with lead time, not an instant integration.
- Multi-tenant white-labeling later implies workspace-level data isolation must be correct from day one (not bolted on) — this drives the Postgres Row-Level-Security-based architecture already in place (Section 19/23).

6. Out of Scope / Non-Goals

Explicitly excluded from this system, to prevent scope creep during development:

- Creating, editing, launching, or pausing live ad campaigns (this is a read-only reporting layer over Meta's Ads Manager, not a replacement for it).
- Creative design or asset production tools (the system displays creative performance, not a creative editor).
- Full CRM functionality (lead assignment, sales pipeline management, calling) — the system consumes lead/CRM data for analytics, it does not replace a CRM like Ashiyana.
- Ad platforms other than Meta (Google Ads, LinkedIn Ads, etc.) at launch — the architecture should not preclude this later, but it is not built now.
- Automated bidding, budget changes, or campaign actions triggered by the system (all recommendations are advisory; a human takes the action in Ads Manager).
- Payment processing / billing for white-label clients — this PRD covers the product itself, not the commercial SaaS billing layer (that would be a separate workstream once white-labeling is prioritized).

7. User Roles

Roles are implemented as a role/permission table pair, not a fixed enum: `workspace_member` links a user to a workspace via a `role_id`, role rows with `workspace_id = null` are system templates available everywhere, and a permission table (`role_id`, resource, action) grants the actual capabilities. Five system templates exist today — CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator — and a workspace can additionally define its own custom roles from the same resource/action primitives. Two earlier templates, Supervisor and Campaign Executive, were removed in migration 0007 and are not covered below.

Note: Because permissions are resource/action rows rather than a hardcoded matrix, the Permissions described per role below reflect the intended default grant for each system template, not a structural limit — a workspace could reconfigure any template's permission rows, or build a custom role that mixes and matches. Section 21 documents the default grant as currently seeded.

7.1 CEO / Business Owner

Purpose: Give the person ultimately accountable for the business a 30-second read on marketing health without needing to ask anyone.

Advantage: Removes the founder as a bottleneck for status updates and replaces gut-feel with a real number every morning.

Attribute	Detail
Goals	Understand overall marketing ROI and spend efficiency at a glance; make go/no-go budget calls.
Responsibilities	Approve major budget shifts; hold managers accountable; review weekly/monthly trends.
Key KPIs Watched	Total spend, estimated ROI, cost per lead, top/bottom performing properties, MoM trend.
Dashboards	CEO Dashboard (Section 11.1).
Permissions	View all workspaces/accounts (full read access), export, approve budget-shift recommendations. No edit access to raw campaign data, no role management.
Reports Received	Daily digest (WhatsApp/email), Monthly Executive Summary.

7.2 Marketing Director

Purpose: Give the person owning marketing outcomes end-to-end a way to manage the whole account portfolio, not just one campaign at a time.

Advantage: Lets one person spot cross-account patterns (e.g. a city underperforming everywhere) that would be invisible looking account-by-account in Ads Manager.

Attribute	Detail
Goals	Optimize budget allocation across accounts and properties; set targets.
Responsibilities	Own the daily/weekly review cadence; set and adjust daily/weekly/monthly targets; escalate account-level issues to Meta support when needed.
Key KPIs Watched	ROAS/ROI, budget pacing, account health, target achievement %, profitability advisor recommendations.
Dashboards	Management / Marketing Director Dashboard (Section 11.2).
Permissions	Full read access across all workspaces they're assigned to; edit targets; approve alert thresholds; export all reports.
Reports Received	Daily digest, Weekly Performance Report.

7.3 Marketing Manager

Purpose: Give the person running day-to-day campaign strategy the operational detail needed to act, not just observe.

Advantage: Cuts the time spent manually pulling Ads Manager exports before every status meeting.

Attribute	Detail
Goals	Hit lead/CPL targets across the workspace; decide which campaigns to keep funding.
Responsibilities	Daily campaign monitoring across the full workspace; bulk-tag campaigns to Property/City (Section 9.2); act on Continue/Pause recommendations from the Profitability Advisor (Section 9.10).
Key KPIs Watched	CPL, CTR, budget utilization, learning-limited campaigns, Estimated Profit/Loss, Continue/Pause status.
Dashboards	Manager Dashboard (Section 11.3).
Permissions	Full read/edit across the workspace's ad accounts and campaigns — no scoping to a subset of accounts. No role management or workspace settings access.
Reports Received	Daily digest, Weekly Performance Report.

7.4 Data Analyst

Purpose: Give the person doing deeper analysis (cohort trends, audience/creative testing) unrestricted access to filter, slice, and export.

Advantage: Avoids analysts having to request raw exports from engineering every time a new question comes up.

Attribute	Detail
Goals	Identify trends across properties, audiences, and creatives that inform strategy; build ad-hoc comparisons.
Responsibilities	Build custom reports; run comparison views; support the Director with data for planning.
Key KPIs Watched	All KPIs, at full historical depth, with drill-down and pivot access.
Dashboards	All dashboards (read-only across workspace), Export Centre, Report Builder.
Permissions	Full read + export access across the workspace; no edit access to live data or settings.
Reports Received	On-demand — this role pulls, rather than receives, reports.

7.5 Administrator (Workspace Admin)

Purpose: Give one person the ability to configure the system itself — users, roles, integrations — without needing engineering involvement for routine changes.

Advantage: Makes the system self-serve for onboarding a new workspace/client, which is essential once white-labeling begins.

Attribute	Detail
Goals	Keep the system correctly configured, secure, and connected to the right Meta assets.
Responsibilities	Manage users and roles (including defining custom roles via the role/permission tables); connect/reconnect Meta Business Manager and ad accounts; configure alert thresholds and notification channels; review audit logs.
Key KPIs Watched	System health metrics: sync success rate, API error rate, active user count.
Dashboards	Admin Panel (Section 20).
Permissions	Full system access: user management, role and permission management, integration settings, audit log access, workspace settings. No implicit access to business KPI dashboards unless also assigned a business role.
Reports Received	System Health Report (weekly).

8. User Stories

A representative set per role — the full backlog will expand these during sprint planning, but every functional requirement in Section 9 traces back to one of these.

Role	User Story
CEO	As a CEO, I want a single number for today's estimated ROI across all accounts, so I don't have to ask anyone for a status update.
Marketing Director	As a Marketing Director, I want to see which of our cities is underperforming this week vs last week, so I can reallocate budget before month-end.
Marketing Manager	As a Marketing Manager, I want to see which campaigns are Learning Limited across the whole workspace, so I can fix them before they waste further spend.
Marketing Manager	As a Marketing Manager, I want the system to flag which campaigns are losing money and recommend pausing them, so I don't have to manually work out profitability for 300+ campaigns.
Data Analyst	As an Analyst, I want to compare creative performance across all properties in a date range, so I can recommend what to brief next.
Administrator	As an Admin, I want to add a new Ad Account to an existing workspace in under 5 minutes, so onboarding a new client doesn't need engineering.

9. Functional Requirements — Modules

Every module below states its Purpose (why we're building it) and Advantage (what it unlocks), followed by concrete requirements, and is tagged MVP or Phase 2 per the phasing decision in Section 24. Build-status notes (in brackets) reflect what's already live in the codebase as of this revision.

9.1 Dashboard (role-based landing page) [MVP — in progress]

Purpose: Give every role a single first screen that answers their top questions without clicking anywhere.

Advantage: Reduces time-to-insight from minutes of manual filtering to seconds on login.

- Role-aware landing page — content differs by role (Section 11): CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator.
- Today vs Yesterday delta on headline KPIs (spend, leads, CPL).
- Top 3 alerts requiring attention, pinned at the top.
- Quick filters: date range, account, property.

9.2 Campaign Monitoring [MVP]

Purpose: Give a live, sortable list of every campaign with health status at a glance.

Advantage: Replaces manually scrolling Ads Manager's 300+ campaign list with a sortable, filterable, taggable table.

- Table view: campaign name, account, status, objective, spend, results, CPL/CPA, delivery status, learning status.
- Health indicator (Green/Amber/Red) computed from CTR, frequency, and pacing thresholds.
- Bulk tagging: manually assign Property and City to campaigns, in bulk or one at a time — no automatic parsing of campaign names (tagging is a permission grant on the Campaign resource, held by Marketing Director and Marketing Manager by default).
- Search and multi-filter combination (see Section 14).

9.3 Campaign Performance (detail view) [MVP]

Purpose: Give a single campaign's full story — trend, ad sets, ads, creatives — in one drill-down screen.

Advantage: Removes the need to jump between Ads Manager tabs to understand one campaign's full picture.

- Time-series trend chart for spend, CTR, CPL over the selected date range.
- Ad set and ad-level breakdown table.
- Creative thumbnails with performance overlay.
- Day-over-day and week-over-week comparison toggle.

9.4 Lead Analytics [MVP]

Purpose: Connect ad spend to the leads it actually produced, regardless of whether they came via Meta Lead Forms or a landing page.

Advantage: This is the single most requested business answer ("which campaign generated maximum leads") — without it, spend and outcome stay disconnected.

- Unified lead table merging Meta Lead Ads data and landing-page/CRM-sourced leads (dual ingestion per Section 5.1).

- Lead volume trend by campaign, property, city, source.
- Cost per Lead (CPL) computed per campaign/property/city.
- Lead quality tagging field (Phase 2: auto-scored; MVP: manual tag synced from CRM where available).

9.5 Property Analytics [MVP]

Purpose: Roll campaign performance up to the property level — the unit real estate leadership actually thinks in.

Advantage: Lets the team answer "which property generated maximum leads" directly, instead of mentally aggregating campaign names.

- Property-level summary: total spend, leads, CPL, estimated ROI.
- Property comparison view (side-by-side, up to 5 properties).
- Property leaderboard, sortable by any KPI.

9.6 Audience Analytics [Phase 2]

Purpose: Show which audience segments are actually converting, not just which have the most reach.

Advantage: Improves targeting efficiency over time by making audience fatigue and winners visible, not buried in Ads Manager breakdowns.

- Audience performance table: age, gender, placement, device breakdowns.
- Audience overlap / fatigue indicator.
- Best-performing audience recommendation per property type.

9.7 Creative Analytics [Phase 2]

Purpose: Show which ad creative (image/video/copy) is actually driving results, and flag fatigue before it hurts CPL.

Advantage: Gives the creative team a data-backed brief for what to make next, instead of guessing.

- Creative-level performance table with thumbnails: CTR, hook rate, cost per view, CPL.
- Creative fatigue alert (frequency rising + CTR declining).
- Creative leaderboard by property/campaign objective.

9.8 Budget Tracking & Pacing [MVP]

Purpose: Show, in real time, whether each campaign/account is on pace to over- or under-spend its budget.

Advantage: Prevents the two most common budget failures: running out mid-month, or under-spending an approved budget.

- Budget utilization % and pacing indicator (ahead/on-track/behind) per campaign and account.
- Forecast spend (simple linear projection at MVP; Phase 2: seasonality-aware forecast).
- Remaining budget and days-until-exhaustion estimate.

9.9 Daily / Weekly / Monthly Reports [MVP]

Purpose: Automatically generate the reports people currently build by hand.

Advantage: Frees the team from manual report-building and guarantees consistent, comparable reporting every period.

- Auto-generated Daily Digest (per role) delivered via email + WhatsApp + in-app.
- Weekly Performance Report (Monday morning, prior week summary).
- Monthly Summary (first business day of the month).
- Phase 2: Quarterly/Yearly and fully custom date-range reports, scheduled report builder.

9.10 Campaign Profitability & Continue/Pause Advisor [MVP]

Purpose: *Turn spend and lead data into a direct, per-campaign answer to "is this making money, and should it keep running" — computed the same way every time, not eyeballed.*

Advantage: *Replaces gut-feel pause/continue calls with a repeatable, auditable rule, and surfaces loss-making campaigns before more budget is wasted on them.*

- Per-campaign inputs: Spend (to date and trailing 7-day), Cost per Lead, Estimated Revenue (assumed conversion % $\times$ average deal value, per Section 5.1), Estimated Profit/Loss = Estimated Revenue – Spend, and trend direction over the trailing 7/14 days.
- Deterministic rule-based classification per campaign: Profitable / Break-even / Loss-making, against workspace-configurable thresholds (e.g. Estimated Profit/Loss margin, days spent below break-even).
- Recommendation output: Continue / Monitor / Reduce Budget / Pause, each paired with a plain-language reason built from the specific rule that fired — not an AI-generated explanation, a templated one traceable to the exact numbers.
- Recommendation is advisory only — no automatic pausing; a human acts on it in Ads Manager, consistent with Section 6 (Out of Scope).
- Thresholds are configurable per workspace by an Admin, not hardcoded.
- Surfaced on the Campaign Monitoring table (Section 9.2), Campaign Detail view (Section 9.3), and as a dedicated widget on the Marketing Director and Marketing Manager dashboards (Section 11). No external AI/LLM API dependency.

9.11 Alerts & Notifications [MVP]

Purpose: *Proactively surface problems the moment they cross a defined threshold, instead of waiting for someone to notice.*

Advantage: *Converts the system from a passive report into an active early-warning system — this is the highest-leverage module in the whole platform.*

- Rule engine per Section 17, configurable thresholds per workspace.
- Delivery via Email, WhatsApp Business API, and in-app push (PWA) as decided in Section 5.
- Acknowledge / Resolve / Escalate workflow with audit trail.

9.12 Export Centre [MVP]

Purpose: *Let any authorized user get their data out, in the format they need, without asking engineering.*

Advantage: *Removes a common support-ticket category ("can you send me this as Excel") entirely.*

- Export to CSV/XLSX for any table view, respecting current filters.
- PDF export for report views (branded, workspace-logo aware for white-label).
- Export history/audit log.

9.13 Admin & Role Management [MVP — live]

Purpose: *Let an Administrator manage users, roles, and permissions without engineering involvement.*

Advantage: *Essential for onboarding new workspaces quickly once white-labeling starts.*

- User invite/deactivate flow (workspace_member table, joining a user to a workspace with a role_id).
- Role assignment per the RBAC matrix (Section 21) — role stored via the role table (system templates: CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator; workspace_id = null marks a template) and a permission table (role_id, resource, action) that also supports defining custom, workspace-specific roles.
- Workspace-level settings: branding (white-label), timezone, currency, alert_thresholds and profitability_thresholds (jsonb).

9.14 Meta API Integration [MVP — live]

Purpose: *Reliably pull campaign, ad set, ad, and insight data from Meta's Marketing API on a schedule.*

Advantage: *This is the data foundation everything else depends on — without reliable sync, every dashboard is stale or wrong.*

- System User-based authentication per workspace — implemented as the pulsecastsyncbot System User on the "PulseCast Ads Sync" Facebook app (Employee-level access, not a personal login).
- Daily scheduled sync via Vercel Cron, plus on-demand manual refresh.
- Sync status visibility (last synced timestamp, error state) surfaced to Admins.
- Graceful handling of Meta API rate limits, token expiry, and partial failures (retry with backoff).

9.15 Audit Logs [MVP]

Purpose: *Record who did what, when — every settings change, alert resolution, and export.*

Advantage: *Required for accountability once multiple people and multiple client workspaces share the system, and for diagnosing "who changed this threshold" disputes.*

- Immutable log of user actions: logins, exports, settings changes, alert acknowledgments, role changes.
- Admin-only searchable log view with date/user/action filters.

9.16 Settings / Workspace Management [MVP — live]

Purpose: *Let each workspace (client) configure its own accounts, users, and preferences independently.*

Advantage: *This is what makes the multi-tenant architecture actually usable by non-technical admins, not just correct in the database.*

- Workspace switcher for users who belong to multiple workspaces (e.g. the founder across real estate + future clients).
- Per-workspace: connected ad accounts, currency, timezone, branding, alert thresholds, notification channel preferences.

10. Non-Functional Requirements

Category	Requirement
Performance	Dashboard initial load under 2s for cached data; filtered queries under 3s at 300+ campaign scale.
Scalability	Data model and sync jobs must handle 10x campaign volume (3,000+) and multiple workspaces without redesign.
Caching	Aggregate KPI rollups pre-computed on sync (not calculated live on every dashboard load).
Offline support (PWA)	Last-synced dashboard data viewable offline; clear "stale data" indicator when offline.
Accessibility	WCAG 2.1 AA color contrast on all charts and KPI cards; keyboard-navigable tables.
Security	Row-Level Security enforced at the database layer per workspace, resolving permissions via the role/permission tables rather than application-layer checks alone; encrypted secrets for Meta API tokens.
Auditability	Every settings change and data export logged (Section 9.15).
Availability	Target 99.5% uptime for the dashboard; sync jobs must alert Admins on failure, not fail silently.
Error handling	User-facing errors are plain-language; technical detail logged, not shown to end users.
Monitoring	Sync job success/failure, API error rates, and page load times tracked and visible to Admins.

11. Dashboard Specifications

Four role-based dashboards cover the five system-template roles (Marketing Director and Marketing Manager dashboards differ mainly in breadth, not structure; both see the full workspace, not a scoped subset. Administrator's primary surface is the Admin Panel per Section 9.13, plus read access to whichever business dashboard it needs).

11.1 CEO Dashboard

Purpose: Answer "is the business's marketing healthy" in one screen, in business language, not ad jargon.

Advantage: The only dashboard the CEO ever needs to open — no training required to read it.

Widgets / Cards: Total spend (MTD/today), Estimated ROI, Estimated Revenue, Top 3 / Bottom 3 properties, Spend trend line (30-day), Alert summary panel, Profitability snapshot (Profitable / Break-even / Loss-making campaign counts).

Filters & Drill-down: Date range, workspace (if multi-client access). No campaign-level drill-down — links out to the Management Dashboard for detail.

11.2 Management / Marketing Director Dashboard

Purpose: Give the person who owns marketing outcomes the full portfolio view with drill-down into any dimension.

Advantage: Lets one person manage 300+ campaigns' worth of complexity without needing to open Ads Manager.

Widgets / Cards: Spend & leads trend (with day-over-day delta), Account health heatmap, Property leaderboard, City leaderboard, Profitability & Continue/Pause recommendation panel (Section 9.10), Budget pacing table, Alert panel, Target achievement gauge (daily/weekly/monthly).

Filters & Drill-down: Date range, account, property, city. Drill-down from any chart into Campaign Monitoring.

11.3 Manager Dashboard

Purpose: Give a Marketing Manager everything needed to run the full campaign portfolio day-to-day.

Advantage: Turns the daily routine from a manual Ads Manager sweep into a 5-minute dashboard review.

Widgets / Cards: Spend/leads/CPL today vs yesterday (full workspace), Campaigns needing attention (Red/Amber), Continue/Pause recommendations (Section 9.10) with reason, Budget pacing, Alert panel.

Filters & Drill-down: Date range, account, property. Drill-down to Campaign Performance detail.

11.4 Analyst / Data Analyst View

Purpose: Give the Data Analyst role unrestricted pivot and comparison tools across the full workspace.

Advantage: Supports ad-hoc questions leadership asks that don't fit a pre-built dashboard.

Widgets / Cards: Pivot table builder, multi-property/multi-campaign comparison view, full KPI library access, saved custom views.

Filters & Drill-down: All filters in Section 14 available in combination; no scope restriction within the workspace.

12. Screen-by-Screen Requirements

Screen	Key Requirements
Login / Workspace Switcher	Email/password or SSO login; if user belongs to >1 workspace, workspace picker shown before dashboard.
Dashboard (role-based)	As specified per role in Section 11; loads in under 2s from cache.
Campaign Monitoring (list)	Sortable/filterable table, health indicators, bulk tagging (permission-gated per role), saved filter views.
Campaign Detail	Trend charts, ad set/ad breakdown, creative thumbnails, comparison toggle.
Lead Analytics	Unified lead table, source breakdown, CPL by dimension, quality tag field.
Property Analytics	Property summary cards, comparison view, leaderboard.
Profitability & Recommendation View	Per-campaign Profit/Loss classification, Continue/Pause recommendation with reason, threshold configuration (Administrator only).
Reports Centre	List of generated reports (daily/weekly/monthly), download/resend, Phase 2 custom report builder.
Alerts Centre	Full alert feed, filter by severity/status, acknowledge/resolve/escalate actions.
Export Centre	Export history, format selector, filter-aware export trigger.
Admin Panel	User management, role and permission assignment, workspace settings, connected ad accounts, audit log.

13. KPI Framework

KPIs are grouped by category rather than listed flat — this makes the (very long) list from the original brief usable in a UI, since each group maps naturally to a filter tab or dashboard section.

13.1 Delivery & Reach

KPI	Definition / Note
Spend	Total amount spent, by campaign/account/day.
Budget & Budget Utilization %	Configured budget vs. amount spent so far.
Impressions	Number of times an ad was shown.
Reach	Number of unique people who saw an ad.
Frequency	$\text{Impressions} \div \text{Reach}$ — high frequency signals ad fatigue.
CPM	Cost per 1,000 impressions.
Delivery Status / Learning Limited	Meta's own delivery health signal, surfaced directly, not recalculated.

13.2 Engagement & Clicks

KPI	Definition / Note
CTR / Unique CTR	Click-through rate, overall and per unique user.
CPC	Cost per click.
Outbound Click / Link Click	Clicks that leave the platform vs. any link click.
Landing Page View	Confirms the click resulted in an actual page load.
Video View / View Rate / Hook Rate	Video-specific engagement depth, key for creative diagnostics.
Post Engagement (Likes, Comments, Shares, Saves)	Organic-style engagement signals on the ad unit.

13.3 Cost Efficiency

KPI	Definition / Note
Cost per Result	Meta's own optimization-event cost metric.
Cost per View / Cost per Purchase	Cost efficiency for video and purchase-objective campaigns.

13.4 Lead & Conversion (core to real estate)

KPI	Definition / Note
Lead / Qualified Lead	Raw lead count vs. CRM-confirmed qualified lead.
CPL (Cost per Lead)	$\text{Spend} \div \text{Leads}$ — the single most-watched KPI for this business.
Conversion Rate	$\text{Leads} \div \text{Clicks}$, or $\text{Qualified Leads} \div \text{Leads}$ depending on the funnel stage viewed.
Lead Quality Score	MVP: manual tag from CRM feedback. Phase 2: modeled score from historical conversion patterns.
Estimated ROAS / Estimated ROI	Computed using assumed conversion % and average deal value per property (Section 5.1) — always labeled "Estimated".

13.5 Budget & Pacing

KPI	Definition / Note
Pacing (Ahead / On-track / Behind)	Spend-to-date vs. expected spend-to-date given the budget period.
Forecast Spend / Forecast Leads	Linear projection at MVP; seasonality-aware in Phase 2.
Remaining Budget	Budget minus spend-to-date.
Daily / Weekly / Monthly Target & Achievement %	User-configured targets vs. actuals.

13.6 Account & Ranking Signals

KPI	Definition / Note
Quality / Engagement / Conversion Ranking	Meta's relative ad-quality diagnostics, surfaced as-is.
Auction Overlap	Flags when your own campaigns compete against each other.

13.7 Profitability & Recommendation (Section 9.10)

KPI	Definition / Note
Estimated Profit/Loss	Estimated Revenue minus Spend for a campaign — the core input to the advisor's classification.
Profitability Classification	Profitable / Break-even / Loss-making, against workspace-configured thresholds.
Recommendation Status	Continue / Monitor / Reduce Budget / Pause — advisory only, always paired with a stated reason.
Days Below Break-even	Consecutive-day counter used by the rule engine before a Pause recommendation triggers.

Phase 2 note: Lead Quality Score and full ROAS become precise once CRM integration is deeper (Section 24); until then they are clearly marked as estimates everywhere they appear in the UI.

14. Filters

Filters are combinable (AND logic across groups, OR within a group — e.g. City: Pune OR Mumbai, AND Recommendation: Pause).

Filter Group	Options
Date	Today, Yesterday, Last 7 Days, Last 30 Days, Custom Range, Week-over-Week / Month-over-Month compare toggle.
Campaign	Campaign, Objective, Status, Buying Type.
Account / Structure	Ad Account, Business Manager (Phase 2, once multi-BM), Platform, Placement, Device.
Geography	Country, State, City.
Business Attribution	Property.
Profitability	Classification (Profitable / Break-even / Loss-making), Recommendation (Continue / Monitor / Reduce Budget / Pause).
Audience & Creative	Audience, Creative, Gender, Age.
Attribution / Tech	Lead Source, Pixel, Event, UTM parameters.

15. Reports

Report	Frequency	Recipients	Phase
Daily Digest	Daily, 9 AM IST	All roles (scoped to their access)	MVP
Weekly Performance Report	Every Monday	Marketing Director, Marketing Manager	MVP
Monthly Executive Summary	1st business day of month	CEO, Marketing Director	MVP
Property Performance Report	Weekly	Marketing Director, CEO	MVP
Campaign Performance Report	On-demand	Marketing Manager	MVP
Profitability & Recommendation Report	Weekly	Marketing Director, Marketing Manager	MVP
Creative Performance Report	Weekly	Marketing Manager, Data Analyst	Phase 2
Audience Performance Report	Weekly	Data Analyst	Phase 2
Lead Quality / Sales Funnel Report	Weekly	Marketing Director, Marketing Manager	Phase 2 (needs deeper CRM link)
Quarterly / Yearly / Custom Report Builder	On-demand	All (permission-gated)	Phase 2

16. Visualizations

Chart choice mapped to what each metric type needs to communicate — not chosen for visual variety.

Metric Type	Recommended Chart
Single headline number (spend, ROI today)	KPI Card with delta indicator (▲/▼vs. yesterday).
Trend over time (spend, CPL, leads)	Line Chart / Area Chart.
Comparison across categories (properties, cities)	Bar Chart / Leaderboard.
Composition (spend by objective, leads by source)	Donut Chart (avoid Pie for >4 segments).
Hierarchical composition (spend by account > campaign > ad set)	Treemap.
Two-dimensional health view (campaign x day)	Heatmap.
Geographic performance (city/state)	Geo Map.
Relationship between two metrics (spend vs. leads)	Scatter Plot / Bubble Chart (bubble size = 3rd metric, e.g. CPL).
Funnel-shaped drop-off (impressions > clicks > leads > sales)	Funnel Chart.
Contribution to a total change (why spend changed WoW)	Waterfall Chart.
Target vs. actual (daily/weekly/monthly target)	Gauge Chart.
Ranked list (top properties, top/bottom profitability)	Leaderboard (styled table, not a generic chart).
Multi-dimensional cross-tab (Analyst use)	Pivot Table / Matrix.

17. Alerts & Notification Rules

Alerts are the highest-leverage module (Section 9.11) — this table is the initial rule set; all thresholds are configurable per workspace by an Admin.

Alert	Trigger Condition	Default Severity	Channel
CTR below threshold	CTR < workspace-defined minimum for 2+ consecutive days	Amber	Email + In-app
CPA / CPL increased	CPL rises >25% vs. trailing 7-day average	Amber	Email + In-app
Budget exhausted	Remaining budget = 0 or campaign paused by Meta for budget	Red	WhatsApp + Email + In-app
Campaign rejected	Meta returns ad review rejection status	Red	WhatsApp + Email + In-app
Campaign stopped unexpectedly	Active campaign transitions to Paused/Inactive without a logged reason	Red	WhatsApp + In-app
Learning Limited	Meta reports Learning Limited delivery status	Amber	In-app
Frequency high	Frequency exceeds workspace threshold (e.g. >3.5 in 7 days)	Amber	In-app
Estimated ROAS dropped	Estimated ROI trend declines >20% week-over-week	Amber	Email + In-app
Pause recommended	Profitability Advisor (Section 9.10) classifies a campaign Loss-making beyond the configured consecutive-day threshold	Red	WhatsApp + Email + In-app
Spend anomaly	Day's spend deviates >40% from trailing 7-day average (either direction)	Red	WhatsApp + Email + In-app
Lead volume dropped	Daily leads fall >30% vs. trailing 7-day average	Amber	Email + In-app
Pixel inactive	No pixel events received for 24+ hours on an active campaign	Red	Email + In-app
Sync / API failure	Meta API sync job fails or token expires	Red (Admin-only)	Email + In-app (Admin)

18. Data Model & ER Diagram

Purpose: Define every entity the system tracks and how they relate, before any table is built.

Advantage: Prevents costly schema rework later — entity relationships defined up front catch data-isolation and migration-ordering issues before they're expensive to fix.

This section is now written directly from the current schema (migrations 0001–0010), not from an idealized design. The full diagram is embedded in Section 18.3.

18.1 Core Entities

- **workspace** — the multi-tenancy boundary; every other entity below belongs to exactly one workspace. Carries name, webhook_secret, alert_thresholds (jsonb), profitability_thresholds (jsonb), timezone, currency.
- **auth.users** — external, managed by Supabase Auth; id and email only. Every user-referencing FK in this schema points here.
- **role** — id, workspace_id (nullable), name, is_system_template. workspace_id = null marks a system template (Section 7); a workspace can also own custom role rows.
- **permission** — id, role_id (FK), resource, action. The grant table behind every role, template or custom.
- **workspace_member** — workspace_id + user_id (composite PK, both FK), role_id (FK). The join between a user, a workspace, and their role.
- **audit_log** — id, workspace_id (FK), user_id (FK, nullable), action, details (jsonb), created_at.
- **business_manager** — id, workspace_id (FK), meta_bm_id, system_user_token_secret_ref, created_at. One per workspace at launch.
- **ad_account** — id, workspace_id (FK), business_manager_id (FK), meta_ad_account_id, name, currency, last_synced_at, last_sync_status, last_sync_error.
- **sync_log** — id, workspace_id (FK), ad_account_id (FK), started_at, finished_at, status, error_message, campaigns_synced, metrics_rows_synced. One row per sync run.
- **pixel** — id, workspace_id (FK), ad_account_id (FK), meta_pixel_id, last_event_at. Tracking asset referenced by landing-page-sourced leads.
- **property** — id, workspace_id (FK), name, city, state, country, assumed_conversion_rate, assumed_avg_deal_value. The real estate unit leads are attributed to; also carries the Section 5.1 revenue assumptions per property.
- **customer** — id, workspace_id (FK), crm_reference, deal_value, closed_at. Populated once CRM integration provides post-lead sales data.
- **campaign** — id, workspace_id (FK), ad_account_id (FK), property_id (FK, nullable), meta_campaign_id, name, objective, status, buying_type, city, tagging_source, daily_budget, lifetime_budget, budget_remaining, budget_synced_at, effective_status, created_at. Note: city is tagged independently of property (bulk-taggable, migration 0008) — it is not derived from the linked property.
- **ad_set** — id, workspace_id (FK), campaign_id (FK), audience_id (FK, nullable), meta_ad_set_id, name.
- **ad** — id, workspace_id (FK), ad_set_id (FK), creative_id (FK, nullable), meta_ad_id.
- **creative** — id, workspace_id (FK), type, thumbnail_url, copy_text.
- **audience** — id, workspace_id (FK), definition (jsonb).
- **daily_metrics** — composite PK (workspace_id, campaign_id, date); spend, impressions, reach, frequency, cpm, ctr, unique_ctr, cpc, clicks, outbound_clicks, landing_page_views, video_views, hook_rate, leads, cpl, budget, budget_utilization_pct, delivery_status, estimated_roi. ctr/cpc/cpm/frequency are Meta's own precomputed values for that single day; clicks (added migration

0010) is a raw summable total, so CTR/CPC can be computed correctly over a date range in SQL rather than by averaging daily ratios.

- `historical_metrics` — same composite PK and column set as `daily_metrics`; the archival copy for data rolled off the hot table (Section 19).
- `profitability_snapshot` — `id`, `workspace_id` (FK), `campaign_id` (FK), `evaluated_at`, `spend_to_date`, `leads_to_date`, `cpl`, `estimated_revenue`, `estimated_profit_loss`, `classification` (`profitable` | `break_even` | `loss_making`), `recommendation` (`continue` | `monitor` | `reduce_budget` | `pause`), `reason`, `days_below_break_even`. One row per campaign per evaluation run, taken at the end of each sync.
- `lead` — `id`, `workspace_id` (FK), `campaign_id` (FK), `property_id` (FK, nullable), `pixel_id` (FK, nullable), `source`, `quality_tag`, `created_at`. Note: `manager_id` / `executive_id` existed in the v1 schema and were dropped in migration 0007 as part of moving to the flat org model this PRD (v4) specifies.
- `alert` — `id`, `workspace_id` (FK), `campaign_id` (FK, nullable), `ad_account_id` (FK, nullable), `rule_key`, `severity`, `status`, `triggered_at`, `resolved_at`, `resolved_by` (FK), `acknowledged_at`, `acknowledged_by` (FK).
- `notification` — `id`, `workspace_id` (FK), `alert_id` (FK, nullable), `report_id`, `user_id` (FK), `channel`, `sent_at`, `read_at`. Delivery record of an Alert or a Report to a user via a channel.

18.2 Key Relationships

- `workspace` 1—* `business_manager` 1—* `ad_account` 1—* `campaign` 1—* `ad_set` 1—* `ad` *—1 `creative`
- `campaign` *—0..1 `property` (nullable — tagging is manual, per Section 5.1)
- `campaign` 1—* `daily_metrics` (time-series fact table), `campaign` 1—* `historical_metrics` (archival)
- `campaign` 1—* `profitability_snapshot` (one per evaluation run)
- `campaign` 1—* `lead`, `lead` *—0..1 `property`, `lead` *—0..1 `customer` (once CRM-linked), `lead` *—0..1 `pixel`
- `ad_account` 1—* `pixel`, `ad_account` 1—* `sync_log`
- `ad_set` *—0..1 `audience`
- `campaign/ad_account` 1—* `alert`, `alert` 1—* `notification`, `notification` *—1 `auth.users`
- `workspace` 1—* `workspace_member`, `workspace_member` *—1 `auth.users`, `workspace_member` *—1 `role`
- `role` 1—* `permission`; `role.workspace_id` = null → system template (available to all workspaces); `role.workspace_id` = a specific workspace → custom role
- `workspace` 1—* `audit_log`, `audit_log` *—0..1 `auth.users`

18.3 ER Diagram

Current schema, migrations 0001–0010. Rendered via PlantUML from the entity list in Section 18.1 and confirmed against the live schema image. This is the authoritative source for Sections 18.1–18.2 and for Section 19.



19. Database Design

Aspect	Recommendation
Primary/Foreign Keys	UUID primary keys throughout (matches Supabase convention); every child table carries a workspace_id foreign key, even where it could be inferred via a join — this is what makes Row-Level Security simple and fast.
Indexes	Composite index on (workspace_id, campaign_id, date) for daily_metrics — this is the hot query path for every dashboard. Composite index on (workspace_id, user_id) for workspace_member for the same reason on the auth path.
Partitioning	Partition daily_metrics / historical_metrics by month once volume grows past ~6-12 months of 300+ campaigns/day — not needed at MVP launch, but the schema should be partition-ready.
Archiving Strategy	Roll data older than 13 months from daily_metrics into the compressed historical_metrics table on a monthly job; dashboards query historical_metrics transparently for older ranges.
Historical Data Strategy	Preserve raw daily granularity for 24 months minimum (real estate sales cycles are long, per background context); aggregate-only beyond that.
Multi-tenancy Enforcement	Postgres Row-Level Security policy on every table keyed to workspace_id, so a bug in application code cannot leak one workspace's data into another's.
Role & Permission Enforcement	role (id, workspace_id nullable, name, is_system_template) and permission (id, role_id, resource, action) tables, joined through workspace_member (workspace_id, user_id, role_id). A null workspace_id on a role row marks it a system template (Section 7); non-null marks a workspace-specific custom role. This is a data-driven grant model, not a fixed enum — RLS and application-layer checks should resolve permissions by querying permission for the caller's role_id, not by switching on a hardcoded role name.
Audit & Sync	audit_log (workspace_id, user_id nullable, action, details jsonb, created_at) records user actions. sync_log (workspace_id, ad_account_id, started_at, finished_at, status, error_message, campaigns_synced, metrics_rows_synced) records each Meta API sync run independently of audit_log.

20. API Design

Internal REST API (consumed by the Next.js frontend and any future white-label client apps), all routes scoped under `/api/workspaces/{workspaceId}/...` to keep the multi-tenancy boundary explicit at the URL level.

Domain	Representative Endpoints
Dashboard	GET /dashboard/summary, GET /dashboard/alerts
Campaign	GET /campaigns, GET /campaigns/{id}, PATCH /campaigns/{id}/tags
Metrics	GET /metrics/daily, GET /metrics/compare
Profitability	GET /campaigns/{id}/profitability, GET /profitability/recommendations, PATCH /workspace/profitability-thresholds
Reports	GET /reports, GET /reports/{id}, POST /reports/generate
Filters	GET /filters/options (returns valid filter values for current workspace)
Export	POST /export, GET /export/history
Authentication	POST /auth/login, POST /auth/refresh, POST /auth/logout
Notification	GET /notifications, PATCH /notifications/{id}/read
Alert	GET /alerts, PATCH /alerts/{id}/acknowledge, PATCH /alerts/{id}/resolve
Roles & Permissions	GET /workspace/roles, POST /workspace/roles (custom role), GET/PATCH /workspace/roles/{id}/permissions
Settings	GET/PATCH /workspace/settings, GET/POST /workspace/users, GET/POST /workspace/ad-accounts

21. Role-Based Access Control (RBAC) Matrix

The five system-template roles as currently seeded, expressed as their default permission grant. Because permissions are stored as (role_id, resource, action) rows rather than a hardcoded switch, a workspace Administrator can adjust any of these or define an additional custom role.

Role	View	Edit (tags/targets)	Delete	Export	Approve	Manage Users/Settings
CEO / Business Owner	All (own workspace)	No	No	Yes	Yes (budget shifts)	No
Marketing Director	All (own workspace)	Yes	No	Yes	Yes	No
Marketing Manager	All (own workspace)	Yes	No	Yes	No	No
Data Analyst	All (own workspace)	No	No	Yes	No	No
Administrator	All (system-wide config)	Settings only	Users only	Yes	No	Yes

22. UI/UX Guidelines

Visual and interaction reference points: Power BI and Looker Studio for dashboard density and drill-down patterns; Grafana for alert-panel styling; Meta Ads Manager for table/filter conventions users already know; HubSpot for the overall approachable, non-technical tone appropriate for CEO/Director personas.

- Minimal, professional visual language — data density without clutter; generous whitespace around KPI cards.
- Responsive: full desktop dashboard experience, plus a genuinely usable mobile/PWA view for on-the-go checks (this is a PWA by requirement, not an afterthought).
- Fast: skeleton loaders, not blank screens, while data loads; cached last-known values shown instantly, live data streamed in after.
- Dark/Light mode, respecting system preference by default.
- Consistent health-status color coding across the entire app: Green = healthy, Amber = needs attention, Red = critical — used identically on every screen, not redefined per module.
- White-label readiness: logo, primary color, and workspace name are theme variables, not hardcoded, from the first UI component built.

23. Technical Architecture

Purpose: *Recommend a stack that's fast to build, well suited to this scale, and ready for multi-tenancy from day one.*

Advantage: *Choosing a proven, well-documented stack up front reduces build risk and keeps the team from re-deciding architecture mid-project.*

Layer	Recommendation	Status
Frontend	Next.js 14 (App Router), TypeScript, Tailwind CSS	Live
Backend	Next.js API routes / server actions (modular monolith, not microservices)	Live
Database	Supabase (managed Postgres) with Row-Level Security per workspace — project ghtonvmmwdpcwzbyegnc	Live
Caching	Pre-computed daily rollup tables (populated by the sync job) + Next.js data caching	In progress
Authentication	Supabase Auth, workspace-scoped sessions with a workspace switcher	Live
Meta Marketing API Integration	System User pulsecastsyncbot per workspace (Employee role), on the "PulseCast Ads Sync" app (App ID 138703756861852), scheduled via Vercel Cron	Live — 5-campaign sync test passed
Background Jobs / Queue	Vercel Cron for scheduled sync (currently 1:00 AM IST daily); a lightweight queue for alert evaluation, notification dispatch, and Profitability Advisor rule evaluation (Section 9.10)	Sync live; queue not yet built
Profitability Rules Engine	Deterministic threshold logic implemented as a Postgres function or Next.js server-side job — no external AI/LLM API call	Not yet built
Notifications	Email via a transactional provider (e.g. Resend/SendGrid); WhatsApp via WhatsApp Business API (Meta Cloud API) with pre-approved templates; in-app via Supabase Realtime	Not yet built
Hosting	Vercel (frontend + API routes)	Live
CI/CD	GitHub Actions → Vercel preview deployments → production on merge to main	Live

23.1 Current Implementation Note

As of this revision, the following pieces of Section 23's recommended architecture already exist in the live codebase (GitHub: ABSCoder1981), per the current schema's ER diagram (migrations 0001–0010), and this PRD has been reconciled against them rather than describing a parallel idealized design:

- role and permission tables, joined through workspace_member — the single source of truth for Section 7 (Roles) and Section 21 (RBAC Matrix) above. Five system-template roles (workspace_id = null) are seeded: CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator. A workspace can also own custom role rows.
- Two earlier templates, Supervisor and Campaign Executive, were removed in migration 0007.
- manager_id / executive_id on lead existed in the v1 schema and were dropped in the same migration 0007, as part of moving to the flat org model this PRD specifies — no user's visibility is scoped to a subset of leads or campaigns within their workspace.
- Workspace-scoped URL structure (/workspaces/[workspaceId]/...) is chosen over session-based workspace tracking.
- Facebook Developer app "PulseCast Ads Sync" and the pulsecastsyncbot System User are live and have completed a successful sync test; each run is now logged in sync_log.

24. MVP vs. Phase 2 Roadmap

MVP covers core tracking, role-based dashboards, and a rule-based profitability advisor; natural-language AI narratives and predictive analytics are deferred.

Phase	Scope	Excludes
MVP (Phase 1)	Dashboard, Campaign Monitoring, Campaign Performance, Lead Analytics, Property Analytics, Budget Tracking, Daily/Weekly/Monthly Reports (auto-generated only), Campaign Profitability & Continue/Pause Advisor (rule-based), Alerts & Notifications, Export Centre, Admin & Role Management, Meta API Integration, Audit Logs, Workspace Settings.	Audience Analytics, Creative Analytics, AI-generated narrative explanations, custom report builder, quarterly/yearly reports, deep CRM/lead-quality scoring.
Phase 2	Audience Analytics, Creative Analytics, Lead Quality scoring (model-based), custom/scheduled Report Builder, seasonality-aware forecasting, deeper CRM integration for true (not estimated) ROI, optional AI-generated plain-English explanations layered on top of the existing rule-based advisor.	—
Phase 3 (Future)	Multi-Business-Manager support for onboarding external white-label clients, billing layer, additional ad platforms (Google Ads), native mobile apps.	—

Sprint order agreed for the MVP build: Meta API + core data model with RLS → Campaign Monitoring → Lead & Property Analytics → Profitability Advisor → Budget Pacing → Alerts (Email/in-app first, WhatsApp deferred) → role-based dashboard variants.

25. Rollout / Migration Plan

- Stage 1 — Connect one Ad Account first (pilot), validate sync accuracy against Ads Manager's own reported numbers before onboarding the remaining accounts.
- Stage 2 — Onboard all remaining Ad Accounts under the single Business Manager; run the system in parallel with any existing manual reporting for one full week to build trust in the numbers.
- Stage 3 — Switch daily/weekly reporting fully to the automated system; retire manual reports.
- Stage 4 — Roll out role-based dashboards, including the Profitability Advisor, to Marketing Directors and Marketing Managers with a short walkthrough (no formal training module needed at this team size).
- Stage 5 (later) — When the first white-label client is signed, use that onboarding as the test of the multi-tenant workspace model end-to-end before selling it as a repeatable process.

26. Risks

Risk	Impact	Mitigation
Meta API rate limits / breaking changes	Sync failures, stale dashboards	Backoff/retry logic, sync-status monitoring surfaced to Admins, version-pin API calls.
"Estimated ROI" mistaken for confirmed revenue	Bad budget decisions made on unreliable numbers	Persistent "Estimated" labeling and a visible assumptions panel (conversion %, deal value) wherever ROI is shown.
Untagged campaigns (manual tagging skipped or delayed)	Broken property/city attribution, campaigns missing from property-level rollups	Bulk tagging UI (Section 9.2) makes manual tagging fast; data-quality completeness indicator on the dashboard flags untagged campaigns.
Pause recommendation acted on too literally	A campaign paused on an early or noisy signal, losing legitimate long-term leads	Recommendation is always advisory with a visible reason and minimum-spend/minimum-day thresholds before it can fire (Section 9.10); no auto-pausing.
WhatsApp Business API approval delay	Alert channel unavailable at launch	Launch alerts on Email + In-app first; add WhatsApp once templates are approved, without blocking MVP.
Multi-tenant data leakage across workspaces	Severe — client data exposure once white-labelled	Database-layer Row-Level Security (Section 19), never relying on application-layer checks alone.
Role permissions drift from what's documented here	A role's actual permission-table grant diverges from what this PRD describes as the build evolves	Section 21's RBAC Matrix documents the default grant as seeded; any change to a role's permission rows should update this PRD the same day, since the underlying model is data-driven and easy to change without a corresponding spec update.
Scope creep back toward the full 25-section spec at once	Delayed MVP launch	Phase gating (Section 24) enforced in sprint planning; Phase 2/3 items explicitly deferred, not silently absorbed.

27. Glossary

Term	Definition
CPL	Cost per Lead — total spend divided by number of leads generated.
CPA	Cost per Action/Acquisition — cost per defined conversion event.
CTR	Click-Through Rate — clicks divided by impressions.
CPM	Cost per Mille — cost per 1,000 impressions.
ROAS	Return on Ad Spend — revenue generated divided by ad spend.
ROI	Return on Investment — net profit relative to total investment, broader than ROAS.
Hook Rate	Percentage of viewers who watch the first few seconds of a video ad — an early-engagement signal.
Frequency	Average number of times a unique person has seen an ad within a period.
Learning Limited	A Meta delivery status indicating a campaign lacks enough conversion events to optimize effectively.
Pacing	Whether spend-to-date is ahead of, on track with, or behind the expected rate given the budget period.
Workspace	The multi-tenancy boundary in this system — one client/organization's isolated data and users.
System Template Role	A role row with workspace_id = null, available to every workspace by default. Five exist today: CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator.
Custom Role	A role row with a non-null workspace_id, defined by a workspace's Administrator from the same permission (resource, action) primitives as a system template.
System User	A Meta Business Manager identity used for server-to-server API access, not tied to a personal login. Implemented as pulsecastsyncbot.
RLS (Row-Level Security)	A database-layer security mechanism restricting which rows a query can return based on the requesting user's context (here, their workspace and role).
Break-even Threshold	The Estimated Profit/Loss point at which a campaign is judged neither profitable nor loss-making — the boundary the Profitability Advisor's rules are set against.
Profitability Advisor	The rule-based module (Section 9.10) that classifies each campaign as Profitable / Break-even / Loss-making and recommends Continue / Monitor / Reduce Budget / Pause, with a stated reason.

28. Acceptance Criteria (MVP)

- All active campaigns (300+) sync daily with a visible last-synced timestamp; sync failures generate an Administrator alert within 15 minutes, and each sync run is recorded in `sync_log`.
- Every role's dashboard (Section 11) loads with role-correct data scope, verified against the RBAC matrix (Section 21) — no user sees or edits data outside their permission grant.
- Daily Digest is delivered automatically via Email and In-app (WhatsApp once approved) to every active user without manual triggering.
- CPL, spend, and lead counts on the dashboard reconcile with Meta Ads Manager's own reported numbers within an agreed tolerance (e.g. $\pm 2\%$) during the pilot stage (Section 25, Stage 1).
- Estimated ROI is visibly and consistently labeled "Estimated" everywhere it appears, with the underlying assumption (conversion %, deal value) visible on hover/click.
- An Administrator can invite a user, assign one of the five system-template roles (or a custom role), and connect a new Ad Account without engineering assistance.
- All alert rules in Section 17 (including the Profitability Advisor's Pause recommendation) fire correctly against test data and can be acknowledged/resolved by an authorized role.
- The Profitability Advisor's classification and recommendation for a given campaign are reproducible from its stated inputs (Spend, Estimated Revenue, thresholds) — the same numbers always produce the same verdict, and are persisted as a `profitability_snapshot` row.
- Row-Level Security is verified to block cross-workspace data access even via a direct API call (not just hidden in the UI) — including verifying that only a role with the relevant permission can modify `workspace_member` or `workspace` settings.

29. Appendix

29.1 Key Design Decisions Log

Decision	Chosen Direction
System scope	New, standalone system — not an extension of any prior project.
Ad account structure	One Business Manager, multiple Ad Accounts at launch.
Client scope	Real estate only at launch; architected to be white-labelled to other clients/verticals later.
Tenancy model	Multi-tenant (workspace-based) from day one, not retrofitted later.
Campaign volume	Designed for 300+ active campaigns at launch.
Lead data sources	Mixed — both native Meta Lead Forms and landing-page/website leads must be supported.
Campaign naming/tagging	No auto-parsing of campaign names — Property/City tagging is manual, in-app, by design (not a naming-convention fallback). City is tagged independently of Property (migration 0008), not derived from it.
Revenue/ROI data	Partial — ROI is estimated using an editable assumed conversion rate and deal value, always labeled "Estimated."
Alert channels	Email + WhatsApp Business API + in-app push (PWA).
Role model (corrected in v4.2)	Roles are a role/permission table pair, not a fixed enum. Five system templates are seeded: CEO, Marketing Director, Marketing Manager, Data Analyst, Administrator — matching the original v4.0 persona set. Two earlier templates, Supervisor and Campaign Executive, were removed in migration 0007. v4.1's four-role guess (Admin/Manager/Marketing Executive/Viewer) was based on a stale conversation summary and has been fully retracted.
Org hierarchy scope	Flat model — no per-person campaign attribution (manager_id/executive_id dropped from lead in migration 0007); all roles see the full workspace, differing only in edit/admin permissions, not in which campaigns they can see.
Profitability advisor	Rule-based (deterministic) Profit/Loss classification and Continue/Pause recommendation on spend/CPL/estimated-revenue thresholds; no external AI/LLM API dependency at MVP.
Release strategy	MVP first (core tracking, dashboards, alerts, rule-based profitability advisor); Audience/Creative Analytics and AI-generated narrative features deferred to Phase 2.

29.2 Next Steps

1. Review and mark up this draft (section-by-section comments welcome).
2. Break the remaining MVP scope (Section 24) into sprint-sized epics for build planning.
3. Confirm WhatsApp Business API account status/lead time before committing to a launch date for that alert channel.